

CA-Software Assignment -2

Principles in Practice. Draft a one-page scenario where you apply Microservices Architecture & Event-Driven Architecture to a hypothetical e-commerce platform. Outline how SOLID principles could enhance the design. Use bullet points to indicate how DRY & KISS principles can be observed in this context.

Scenario: Hypothetical E-commerce platform

1. Application of Microservices Architecture:

- Each service manages a single business functions:
 - Product service (catalog, pricing)
 - Order service (checkout, tracking)
 - Payment service (transactions)
 - User service (authentication, profile)
 - Inventory service (stock management)
- Benefits:
 - Independent deployment
 - Scalability per service
 - Fault isolation

2. Application of Event-Driven Architecture (EDA).

- Services communicate through events:
 - OrderPlaced, Payment Completed, ItemOutOfStock
- Uses Message brokers like Kafka or RabbitMQ
- Promotes loose coupling & asynchronous communication

3. Applying SOLID principles:

- S (Single Responsibility): Each service handles one concern (eg, Payment handles only payment)
- O (Open/Closed): Services can be extended (eg, adding a new payment gateway) without changing core logic.
- L (Liskov Substitution): Interfaces for logging, payment can be replaced without affecting services.
- I (Interface Segregation): Services expose only necessary APIs.
- D (Dependency Inversion): Use abstractions for services & message brokers.

4. Observing DRY & KISS Principles:

- DRY (Don't Repeat Yourself):
 - Reusable libraries for authentication & logging

- shared event formats for communication
- KISS (keep it simple, stupid):
 - lightweight services
 - Minimal dependencies
 - Clean APIs with REST or gRPC